



Data Article

PM₁₀ Plume dispersion data of the Zouk power plant in Lebanon



Samer Salloum^a, Julie Nassar^b, Rima Baalbaki^b,
Alan L. Shihadeh^a, Najat A. Saliba^b, Issam Lakkis^{a,*}

^a Department of Mechanical Engineering, Faculty of Engineering and Architecture, American University of Beirut, Beirut 1107 2020, Lebanon

^b Department of Chemistry, Faculty of Arts and Sciences, American University of Beirut, Beirut 1107 2020, Lebanon

ARTICLE INFO

Article history:

Received 6 June 2018

Received in revised form

12 September 2018

Accepted 17 September 2018

Available online 19 September 2018

Keywords:

Ambient air pollution

PM₁₀

Zouk Mikael power plant

TAPM

ABSTRACT

Ambient air pollution is a major risk to the human health and to the environment. The data presented quantifies the (PM₁₀) contribution of the Zouk Mikael power plant to the ambient air pollution in Lebanon for the year 2014. The data is the outcome of a computer simulation using The Air Pollution Model (TAPM), taking into account the emission source data, the spatio-temporal meteorological conditions, the terrain height, and the land cover characteristics. The data set presents the annual, seasonal and monthly averages of the spatial distribution of the ground-level particulate (PM₁₀) concentrations in the ambient air. The data set also includes spatial distribution of the maximum concentrations, which revealed two zones of elevated concentrations. Monthly averages and maximum concentrations in these two zones are also reported. Analysis of the data can provide information on the health risk the residents in the affected areas are subjected to. The data can also provide insight on the impact of the meteorological conditions (temperature and velocity) and the topography on pollutant dispersion in regions bounded by the sea and by a mountain range.

© 2018 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

* Corresponding authors.

E-mail address: ilakkis@gmail.com (I. Lakkis).

Specifications table

Subject area	Modeling and simulation
More specific subject area	Air Pollution Modeling
Type of data	Figures and barcharts
How data was acquired	Computer simulation using The Air Pollution Model (TAPM)
Data format	Analyzed
Experimental factors	• The ratio of Total Suspended Particles to PM10 is assumed to be 0.5 [2]. • An ideal scenario of regular maintenance is assumed. • Although the complex operating schedule of the power plant is not taken into account, the average emitted mass flow rate is conserved.
Experimental features	• The Air Pollution Model takes as its input (1) the terrain height data, land cover characterization data, soil texture types, mean sea surface temperatures, (2) six-hourly synoptic scale 4D conditions for the year 2014 and (3) the emission sources locations, dimensions, chemical composition, flowrate, and thermophysical conditions.
Data source location	Zouk Mikael, Lebanon (33.9703°N, 35.6206°E)
Data accessibility	Data is with this article
Related research article	[1] Rima Baalbaki, Julie Nassar, Samer Salloum, Alan L. Shihadeh, Issam Lakkis, Najat A. Saliba, "Comparison of atmospheric polycyclic aromatic hydrocarbon levels in three urban areas in Lebanon", Atmospheric Environment, Volume 179, 2018, Pages 260–267, ISSN 1352–2310, https://doi.org/10.1016/j.atmosenv.2018.02.028 .

Value of the data

- The data could be used to assess the contribution of the power plant to the pollution in the nearby region.
- The data can be used to identify zones of elevated PM₁₀ concentrations.
- The data can be used to decide on distribution and locations of pollution monitoring sensors.
- The data can be used in conjunction with long term measurements to improve pollution forecast for both the short term as well as the long term.
- The data can be used to carry out geographically based health risk assessment.
- The data can be used to decide on sample size and distribution for correlating pollution exposure with incidences of diseases.
- The data can be used by environmental scientists to gain insight into seasonal variations of the pollution dispersion in similar settings.
- The data can be used by researchers in environmental policy to propose changes in exisiting conditions and/or propose changes in the policies for the future.

1. Data

TAPM outputs show the power plant stack emission trajectories for a whole year. An animation can be found through the following link <https://www.youtube.com/watch?v=aml0oELAdBA>.

Annual average PM₁₀ concentration map at ground level is shown in Fig. 1. Seasonal average PM₁₀ concentration mapsat ground level are shown in Fig. 2. Monthly average PM₁₀ concentration maps at ground level are shown in Fig. 3. Fig. 4 shows the map of the maximum annual PM₁₀ concentration,

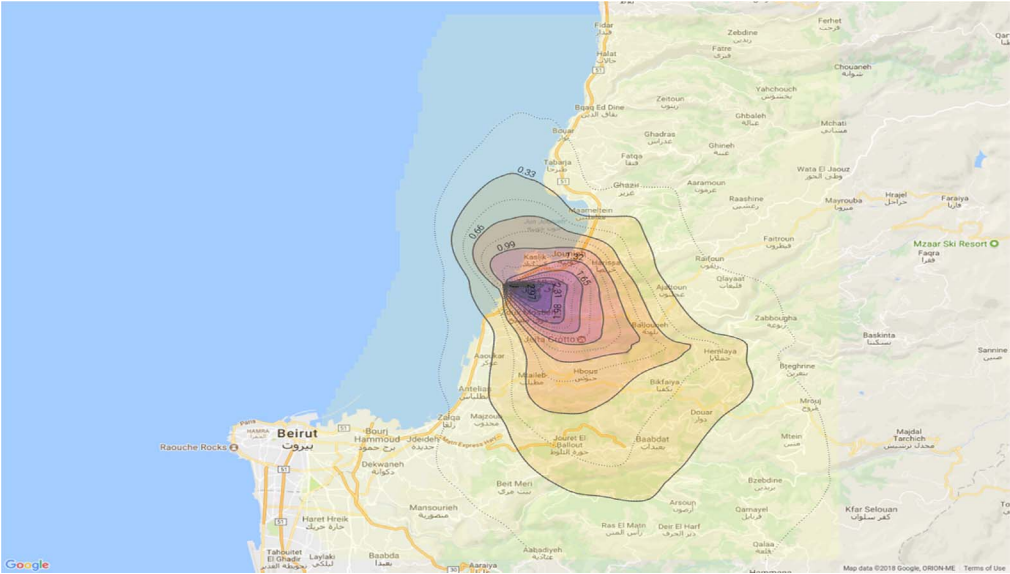


Fig. 1. Average annual PM₁₀ concentration in µg/m³ for the year 2014.

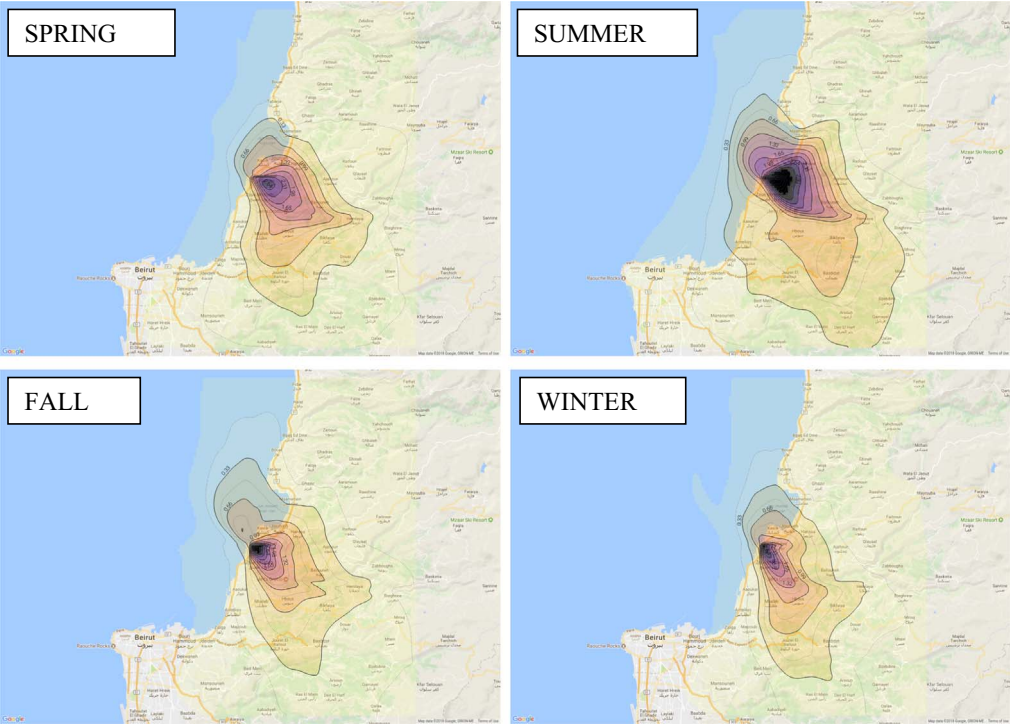


Fig. 2. Average seasonal PM₁₀ concentration in µg/m³ for the year 2014.

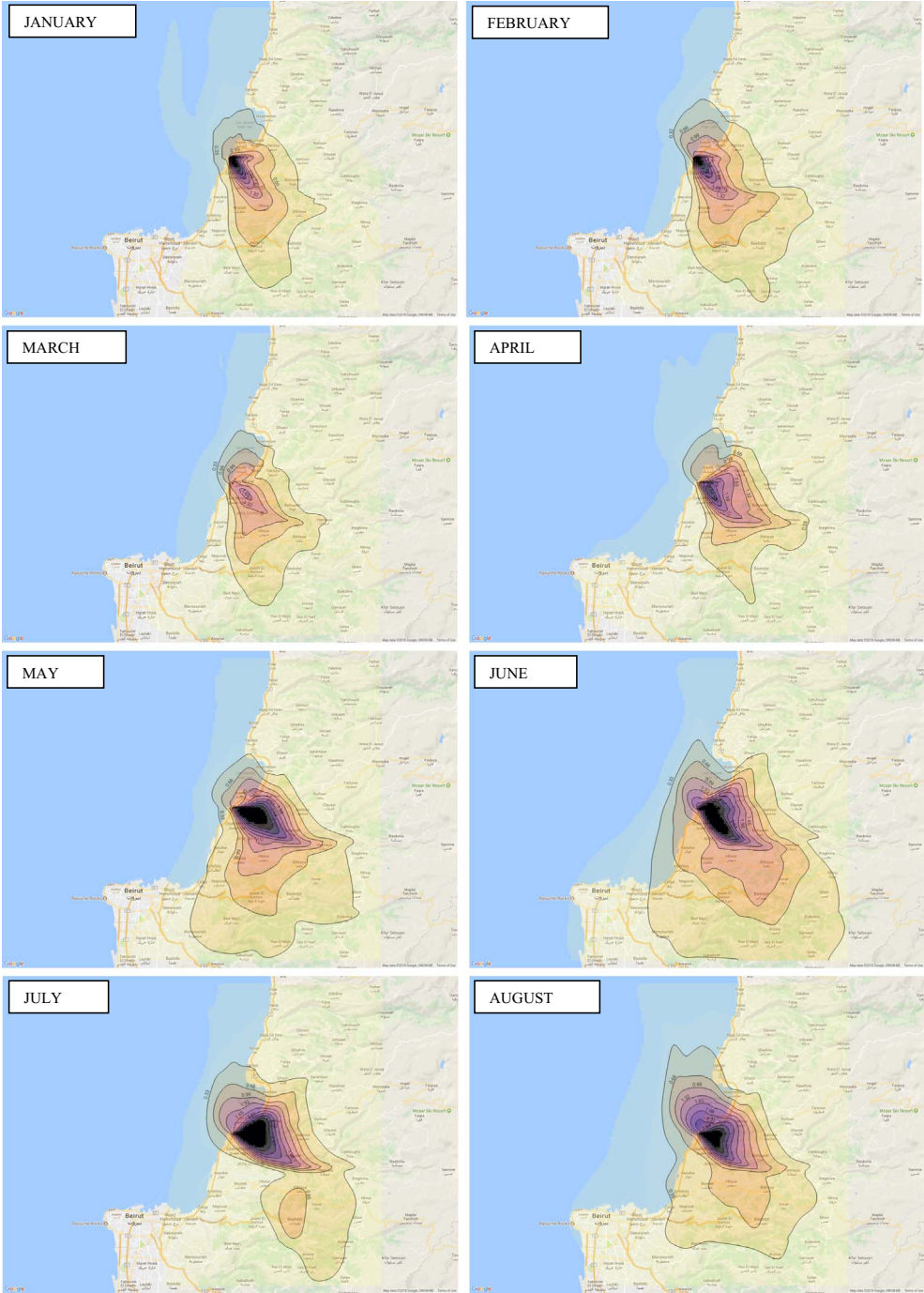


Fig. 3. Average monthly PM₁₀ concentration in $\mu\text{g}/\text{m}^3$ for the year 2014.

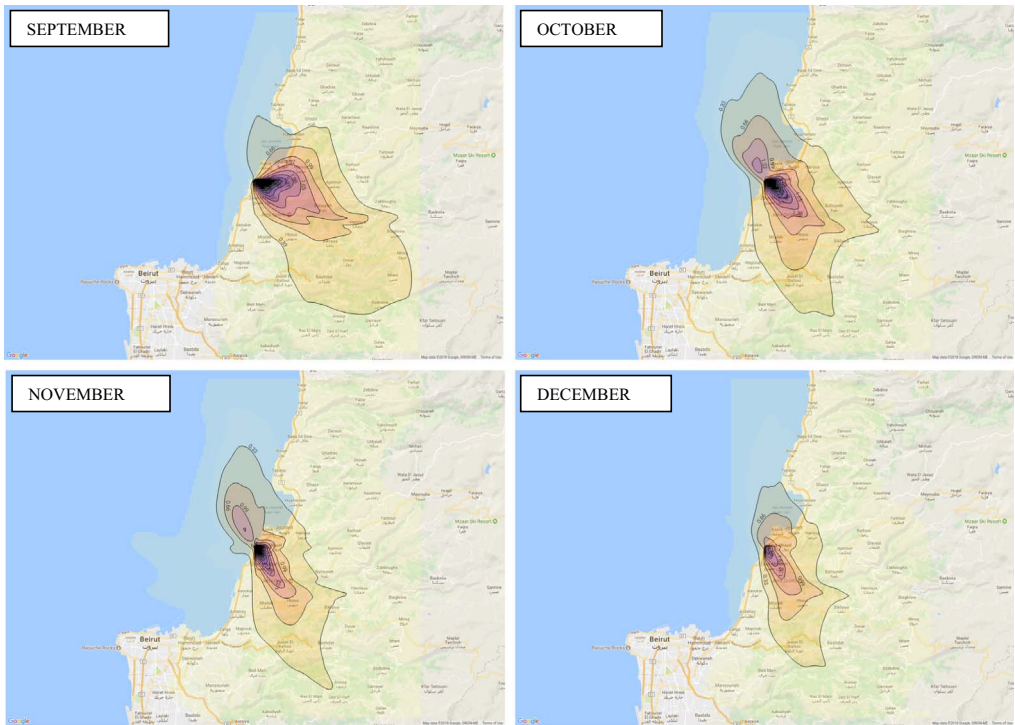


Fig. 3. (continued)

where two distinct zones of elevated maximum concentrations are identified. Fig. 5 presents a plot of monthly variation of the maximum PM_{10} concentration in Zones 1 and 2. Fig. 6 presents a plot of monthly variation of the average PM_{10} concentration in Zones 1 and 2.

2. Experimental design, materials, and methods

PM_{10} emission rate are calculated from data provided by EDL Engineers as described in the Related research article [1]. Geometric characteristics of the plant turbines and stacks as well as the plant running schedule are also detailed in the Related research article [1]. A summary is presented here:

Emissions source:

- four stacks, each stack is 9 m in diameter and 145 m high.
- On average, 2.5 out of 4 stream turbines operate continuously.
- PM_{10} emissions of 10.2 g/s through the stacks at a temperature of 700 K.

The Air Pollution Model [3,4] (TAPM) is a computational tool that predicts pollutant transport, taking into account atmospheric chemical reactions, and species advection by weather flow field patterns [4]. TAPM numerically solves an approximation of the Navier Stokes (N–S) equations to predict the wind velocity field over a prescribed period of time and in a selected region. The wind velocity field is then used as an input to solve the species transport equation in order to obtain the hourly concentration profile for the pollutants. The model, which employs nested grids, starts solving the governing equations on the outermost grid, where the boundary and initial conditions are

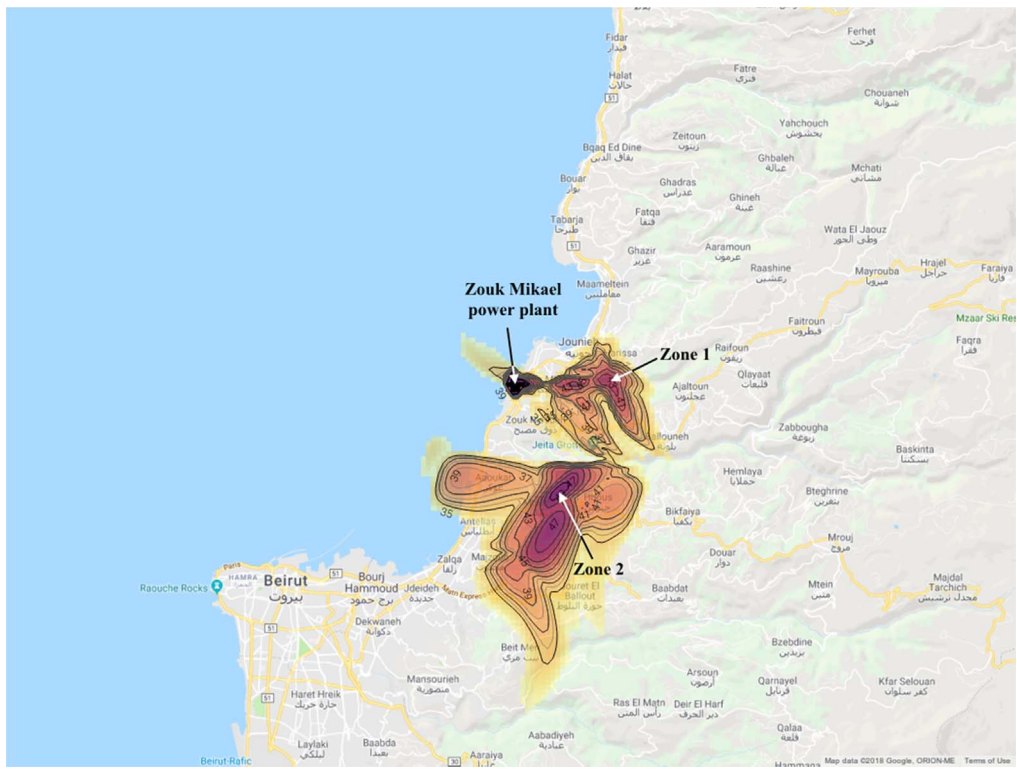


Fig. 4. Maximum annual PM₁₀ concentration in µg/m³ for the year 2014 showing two zones of elevated PM₁₀ concentrations.

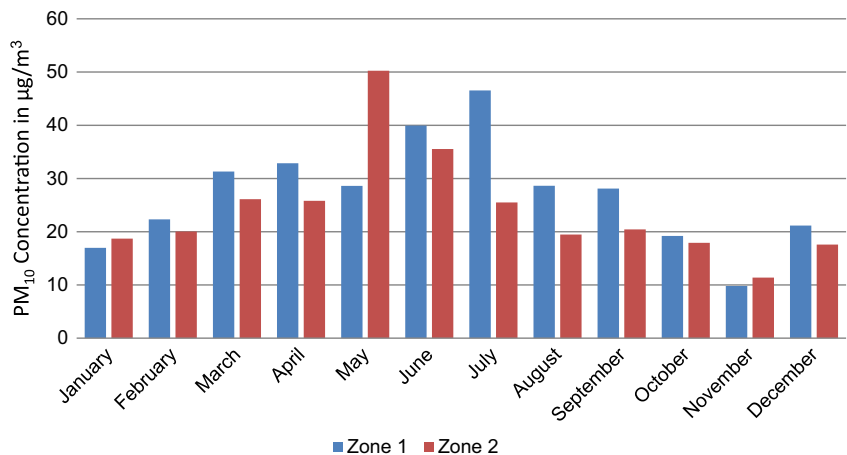


Fig. 5. Maximum PM₁₀ concentration in µg/m³ at Zone 1 and Zone 2 for every month in 2014.

obtained from global synoptic analyses. It then passes the large-scale information as boundary conditions to the next (finer) grid. Radiative fluxes, both at the surface and at upper levels, are also included [4]. The soil temperature and moisture content are calculated following the method presented in [5] using some empirically derived constants or functions of soil moisture content and soil texture type.

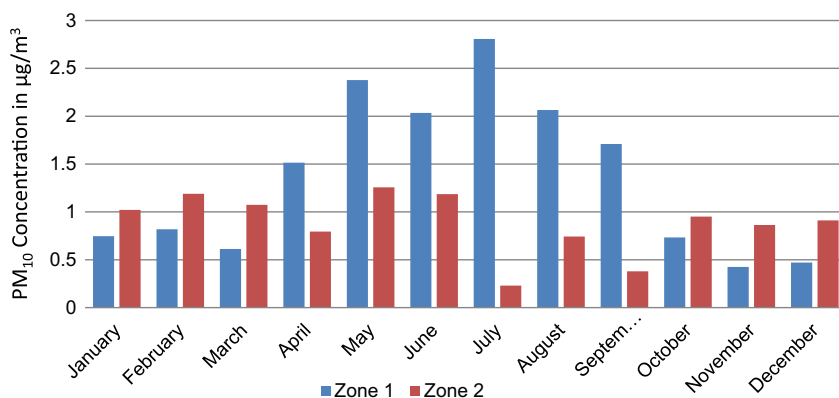


Fig. 6. Average PM₁₀ concentration in µg/m³ at Zone 1 and Zone 2 for every month in 2014.

Acknowledgments

This work was supported in part by the Lebanese National Council for Scientific Research and the Zouk Mikael Municipality.

Transparency document. Supplementary material

Transparency data associated with this article can be found in the online version at <https://doi.org/10.1016/j.dib.2018.09.047>.

References

- [1] Rima Baalbaki, Julie Nassar, Samer Salloum, Alan L. Shihadeh, Issam Lakkis, Najat A. Saliba, Comparison of atmospheric polycyclic aromatic hydrocarbon levels in three urban areas in Lebanon, *Atmos. Environ.* 179 (2018) 260–267.
- [2] H.L. Goldstein, C.W. Siegmund, Influence of heavy fuel oil composition and boiler combustion conditions on particulate emissions, *Environ. Sci. Technol.* 12 (1976) 1109–1114 (10).
- [3] P. Hurley, TAPM V4. Part 1: Technical Description. CSIRO Marine and Atmospheric Research Paper, 2008.
- [4] P.J. Hurley, W.L. Physick, A.K. Luhar, TAPM: a practical approach to prognostic meteorological and air pollution modelling, *Environ. Model. Softw.* 20 (2005) 737–752.
- [5] R. Pielke, *Mesoscale Meteorological Modelling*, Academic Press, 2013.